

INTEGRATION

by substitution

Question 1

Carry out the following integrations **by substitution** only.

$$1. \int 4x(2x-1)^4 dx = \frac{1}{6}(2x-1)^6 + \frac{1}{5}(2x-1)^5 + C$$

$$2. \int \frac{2x}{2x+1} dx = \frac{1}{2}(2x+1) - \frac{1}{2} \ln|2x+1| + C$$

$$3. \int x(4-x^2)^{-\frac{1}{2}} dx = -(4-x^2)^{\frac{1}{2}} + C$$

$$4. \int \frac{4x}{6x^2-1} dx = \frac{1}{3} \ln|6x^2-1| + C$$

$$5. \int x(3x-1)^4 dx = \frac{1}{54}(3x-1)^6 + \frac{1}{45}(3x-1)^5 + C$$

$$6. \int \frac{8x}{\sqrt{4x-1}} dx = \frac{1}{3}(4x-1)^{\frac{3}{2}} + (4x-1)^{\frac{1}{2}} + C$$

$$7. \int \frac{2x^2}{\sqrt{2x^3+1}} dx = \frac{2}{3}(2x^3+1)^{\frac{1}{2}} + C$$

$$8. \int \frac{4-3x}{x+2} dx = 10 \ln|x+2| - 3(x+2) + C$$

$$9. \int \frac{4x^2}{2x-1} dx = \frac{1}{4}(2x-1)^2 + (2x-1) + \frac{1}{2} \ln|2x-1| + C$$

$$10. \int \frac{4x-3}{3x-4} dx = \frac{4}{9}(3x-4) + \frac{7}{9} \ln|3x-4| + C$$

$$\begin{aligned}
 1. \quad & \int 4x(2x-1)^4 dx = \int (4x) u^4 \frac{du}{2} = \int 2x u^4 du \quad \begin{cases} u = 2x-1 \\ \frac{du}{dx} = 2 \\ 2x = u+1 \end{cases} \\
 & = \int (u+1)u^4 du = \int u^5 + u^4 du \\
 & = \frac{1}{6}u^6 + \frac{1}{5}u^5 = \frac{1}{6}(2x-1)^6 + \frac{1}{5}(2x-1)^5 + C \\
 2. \quad & \int \frac{2x}{2x+1} dx = \int \frac{2x}{u} \cdot \frac{du}{2} = \int \frac{u-1}{u} \times \frac{du}{2} \quad \begin{cases} u = 2x+1 \\ \frac{du}{dx} = 2 \\ 2x = u-1 \end{cases} \\
 & = \frac{1}{2} \int \left(1 - \frac{1}{u}\right) du = \frac{1}{2} \left[u - \ln|u|\right] + C \\
 & = \frac{1}{2}(2x+1) - \frac{1}{2}\ln|2x+1| + C \\
 3. \quad & \int 2x(4-3x)^{\frac{3}{2}} dx = \int x u^{\frac{3}{2}} \times \frac{du}{-3} = -\frac{1}{3} \int u^{\frac{3}{2}} du \quad \begin{cases} u = 4-3x \\ \frac{du}{dx} = -3 \\ dx = \frac{du}{-3} \end{cases} \\
 & = -\frac{1}{3} \int u^{\frac{3}{2}} du = -\frac{1}{3} \times \frac{2}{5} u^{\frac{5}{2}} + C = -\frac{2}{15} (4-3x)^{\frac{5}{2}} + C \\
 4. \quad & \int \frac{4x}{x^2-1} dx = \int \frac{4x}{u} \times \frac{du}{2x} = \int \frac{2}{u} du \quad \begin{cases} u = x^2-1 \\ \frac{du}{dx} = 2x \\ dx = \frac{du}{2x} \end{cases} \\
 & = \frac{1}{2} \ln|u| + C = \frac{1}{2} \ln|x^2-1| + C \\
 5. \quad & \int 2(3x-1)^9 dx = \int 2u^9 \times \frac{du}{3} = \frac{2}{3} \int u^9 du \quad \begin{cases} u = 3x-1 \\ \frac{du}{dx} = 3 \\ dx = \frac{du}{3} \end{cases} \\
 & = \frac{2}{3} \times \frac{1}{10} u^{10} + C = \frac{1}{15} (3x-1)^{10} + C \\
 & = \frac{1}{15} \left[6u^5 + \frac{1}{5}u^4 \right] = \frac{1}{15} (3x-1)^5 + \frac{1}{75} (3x-1)^4 + C
 \end{aligned}$$

$$\begin{aligned}
 6. \quad & \int \frac{8x}{x^2(x^2+1)} dx = \int \frac{8x}{x^2} \times \frac{1}{x^2+1} dx = \int \frac{8}{x(x^2+1)} dx \quad \begin{cases} u = x^2+1 \\ \frac{du}{dx} = 2x \\ 2x = u-1 \end{cases} \\
 & = \int \frac{2(4-u)}{u(u-1)} du = \frac{1}{2} \int \frac{4-u}{u(u-1)} du = \frac{1}{2} \int \frac{4-u}{u^2-u} du \\
 & = \frac{1}{2} \int \left(\frac{4}{u^2-u} - \frac{1}{u-1} \right) du = \frac{1}{2} \int \left(\frac{4}{u(u-1)} - \frac{1}{u-1} \right) du \\
 & = \frac{1}{2} \int \left(\frac{4}{u(u-1)} - \frac{1}{u-1} \right) du = \frac{1}{2} \int \left(\frac{4}{u(u-1)} - \frac{1}{u-1} \right) du \\
 7. \quad & \int \frac{2x^2}{x^2(x^2+1)} dx = \int \frac{2x^2}{x^2} \times \frac{1}{x^2+1} dx = \int \frac{2}{x^2+1} dx \quad \begin{cases} u = x^2+1 \\ \frac{du}{dx} = 2x \\ 2x = u-1 \end{cases} \\
 & = \int \frac{2}{u} du = 2 \ln|u| + C = 2 \ln|x^2+1| + C \\
 8. \quad & \int \frac{4-3x}{x^2+1} dx = \int \frac{4-3x}{u} \times \frac{du}{2x} = \frac{1}{2} \int \frac{4-3x}{u} du \quad \begin{cases} u = x^2+1 \\ \frac{du}{dx} = 2x \\ dx = \frac{du}{2x} \end{cases} \\
 & = \frac{1}{2} \int \frac{4-3x}{u} du = \frac{1}{2} \int \frac{4-3x}{u} du = \frac{1}{2} \int \frac{4-3x}{u} du \\
 9. \quad & \int \frac{1}{x^2+1} dx = \int \frac{1}{u} \times \frac{du}{2x} = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln|u| + C \quad \begin{cases} u = x^2+1 \\ \frac{du}{dx} = 2x \\ dx = \frac{du}{2x} \end{cases} \\
 & = \frac{1}{2} \ln|x^2+1| + C \\
 10. \quad & \int \frac{4x-3}{x^2+1} dx = \int \frac{4x-3}{u} \times \frac{du}{2x} = \frac{1}{2} \int \frac{4x-3}{u} du \quad \begin{cases} u = x^2+1 \\ \frac{du}{dx} = 2x \\ dx = \frac{du}{2x} \end{cases} \\
 & = \frac{1}{2} \int \frac{4x-3}{u} du = \frac{1}{2} \int \frac{4x-3}{u} du = \frac{1}{2} \int \frac{4x-3}{u} du \\
 & = \frac{1}{2} \int \left(\frac{4x}{u} - \frac{3}{u} \right) du = \frac{1}{2} \left(4 \ln|u| - \frac{3}{u} \right) + C \\
 & = \frac{1}{2} \left(4 \ln|x^2+1| - \frac{3}{x^2+1} \right) + C
 \end{aligned}$$

Question 2

Carry out the following integrations **by substitution** only.

$$1. \int 6x(3x-1)^3 dx = \frac{2}{15}(3x-1)^5 + \frac{1}{6}(3x-1)^4 + C$$

$$2. \int \frac{5x}{5x-1} dx = \frac{1}{5}(5x-1) + \frac{1}{5} \ln|5x-1| + C$$

$$3. \int 3x(x^2+1)^{\frac{1}{2}} dx = (x^2+1)^{\frac{3}{2}} + C$$

$$4. \int \frac{3x^2}{2x^3+1} dx = \frac{1}{2} \ln|2x^3+1| + C$$

$$5. \int x(2x-1)^5 dx = \frac{1}{24}(2x-1)^6 + \frac{1}{28}(2x-1)^7 + C$$

$$6. \int \frac{10x}{\sqrt{1-2x}} dx = \frac{5}{3}(1-2x)^{\frac{3}{2}} - 5(1-2x)^{\frac{1}{2}} + C$$

$$7. \int \frac{3x^4}{\sqrt{2x^5+1}} dx = \frac{3}{5}(2x^5+1)^{\frac{1}{2}} + C$$

$$8. \int \frac{1-x}{1+2x} dx = \frac{3}{4} \ln|1+2x| - \frac{1}{4}(1+2x) + C$$

$$9. \int \frac{6x^2}{2x+3} dx = \frac{3}{8}(2x+3)^2 - \frac{9}{2}(2x+3) + \frac{27}{4} \ln|2x+3| + C$$

$$10. \int \frac{1}{x^{\frac{1}{2}} \sqrt{x^{\frac{1}{2}}-1}} dx = 4\sqrt{x^{\frac{1}{2}}-1} + C$$

$$\begin{aligned}
 1. \int 6x(3x-1)^5 dx &= \int 6x u^5 \frac{du}{dx} = \frac{1}{2} \int (2u+2) u^5 du = \begin{cases} u = 3x-1 \\ \frac{du}{dx} = 3 \\ dx = \frac{du}{3} \\ 2u = 2(3x-1) \\ 6x = 2u+2 \end{cases} \\
 &= \frac{1}{2} \int (2u+2) u^5 du = \frac{1}{2} \left[\frac{2}{6} u^6 + \frac{2}{6} u^6 \right] = \frac{1}{3} u^6 + \frac{1}{3} u^6 + C \\
 &= \frac{2}{3} (3x-1)^6 + \frac{1}{3} (3x-1)^6 + C \\
 2. \int \frac{5x}{5x^2-1} dx &= \int \frac{\frac{5}{2} \frac{du}{dx}}{\frac{u}{2}} = \int \frac{5}{u} \times \frac{du}{2} = \begin{cases} u = 5x-1 \\ \frac{du}{dx} = 5 \\ dx = \frac{du}{5} \\ 5x = u+1 \end{cases} \\
 &= \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \left[\ln|u| \right] = \frac{1}{2} \ln|5x-1| + C \\
 3. \int 3x(x^2+1)^{\frac{1}{2}} dx &= \int 3x u^{\frac{1}{2}} \frac{du}{dx} = \int \frac{3}{2} u^{\frac{1}{2}} du = \begin{cases} u = x^2+1 \\ \frac{du}{dx} = 2x \\ dx = \frac{du}{2x} \end{cases} \\
 &= \frac{3}{2} \cdot \frac{2}{3} u^{\frac{3}{2}} + C = \frac{1}{2} (x^2+1)^{\frac{3}{2}} + C \\
 4. \int \frac{3x^2}{2x^3+1} dx &= \int \frac{3x^2}{u} \times \frac{du}{dx} = \frac{1}{2} \int \frac{1}{u} du = \begin{cases} u = 2x^3+1 \\ \frac{du}{dx} = 6x^2 \\ dx = \frac{du}{6x^2} \end{cases} \\
 &= \frac{1}{2} \ln|u| + C = \frac{1}{2} \ln|2x^3+1| + C \\
 5. \int x(2x-1)^2 dx &= \int x u^2 \frac{du}{dx} = \int \frac{u^2}{2} \times \frac{du}{dx} = \begin{cases} u = 2x-1 \\ \frac{du}{dx} = 2 \\ dx = \frac{du}{2} \\ u+1 = 2x \end{cases} \\
 &= \frac{1}{2} \int (u+1) u^2 du = \frac{1}{2} \left[\frac{1}{3} u^3 + \frac{1}{2} u^2 \right] = \frac{1}{6} u^3 + \frac{1}{4} u^2 \\
 &= \frac{1}{6} (2x-1)^3 + \frac{1}{4} (2x-1)^2 + C
 \end{aligned}$$

$$\begin{aligned}
 6. \int \frac{10x}{\sqrt{1-2x}} dx &= \int \frac{10x}{u} \times \frac{du}{dx} = \int \frac{5-u}{u} \times \frac{du}{2} = \begin{cases} u = 1-2x \\ \frac{du}{dx} = -2 \\ dx = \frac{du}{-2} \\ 2x = 1-u \\ 10x = 5-5u \end{cases} \\
 &= \frac{1}{2} \int \frac{5-u}{u} du = \frac{1}{2} \int \frac{5}{u} - \frac{u}{u} du = \frac{1}{2} \left[5 \ln|u| - \frac{1}{2} u^2 \right] \\
 &= \frac{5}{2} \ln|1-2x| - \frac{1}{4} (1-2x)^2 + C \\
 7. \int \frac{2x^3}{\sqrt{2x^2+1}} dx &= \int \frac{2x^3}{u} \times \frac{du}{dx} = \int \frac{3}{10} u^{\frac{1}{2}} du = \begin{cases} u = 2x^2+1 \\ \frac{du}{dx} = 4x \\ dx = \frac{du}{4x} \end{cases} \\
 &= \frac{3}{10} \cdot \frac{2}{3} u^{\frac{3}{2}} + C = \frac{2}{15} u^{\frac{3}{2}} + C = \frac{2}{15} (2x^2+1)^{\frac{3}{2}} + C \\
 8. \int \frac{1-2x}{1+2x} dx &= \int \frac{1-u}{u} \times \frac{du}{2} = \frac{1}{2} \int \frac{1-u}{u} du = \begin{cases} u = 1+2x \\ \frac{du}{dx} = 2 \\ dx = \frac{du}{2} \\ 2x = u-1 \end{cases} \\
 &= \frac{1}{2} \int \frac{1-u}{u} du = \frac{1}{2} \left[\ln|u| - \frac{1}{2} u^2 \right] + C = \frac{1}{2} \ln|1+2x| - \frac{1}{4} (1+2x)^2 + C \\
 9. \int \frac{5x^2}{2x^3+3} dx &= \int \frac{3x^2}{u} \times \frac{du}{dx} = \int \frac{3}{2} \frac{u^{\frac{1}{2}}}{u} du = \begin{cases} u = 2x^3+3 \\ \frac{du}{dx} = 6x^2 \\ dx = \frac{du}{6x^2} \end{cases} \\
 &= \int \frac{3x^2}{2x^3+3} du = \int \frac{3x^2}{u} du = \frac{3}{2} \int \frac{1}{u} du = \frac{3}{2} \ln|u| + C \\
 &= \frac{3}{2} \ln|2x^3+3| + C \\
 10. \int \frac{1}{x^2 \sqrt{1-x^2}} dx &= \int \frac{1}{x^2} \times \frac{du}{dx} = \int \frac{2}{u^2} du = \begin{cases} u = x^2-1 \\ \frac{du}{dx} = 2x \\ dx = \frac{du}{2x} \end{cases} \\
 &= \int \frac{2}{u^2} du = -\frac{2}{u} + C = -\frac{2}{x^2-1} + C
 \end{aligned}$$

Question 3

Carry out the following integrations **by substitution** only.

$$1. \int 10x(5x-3)^3 dx = \frac{2}{25}(5x-3)^5 + \frac{3}{10}(5x-3)^4 + C$$

$$2. \int \frac{12x}{2x-1} dx = 3(2x-1) + 3\ln|2x-1| + C$$

$$3. \int x(x^2-1)^{\frac{5}{2}} dx = \frac{1}{7}(x^2-1)^{\frac{7}{2}} + C$$

$$4. \int \frac{5x^5}{2x^6+7} dx = \frac{5}{12}\ln|2x^6+7| + C$$

$$5. \int 2x(1-5x)^4 dx = -\frac{2}{125}(1-5x)^5 + \frac{1}{75}(1-5x)^6 + C$$

$$6. \int \frac{9x}{\sqrt{4x^2+1}} dx = \frac{9}{4}\sqrt{4x^2+1} + C$$

$$7. \int \frac{3x-1}{\sqrt{4x-1}} dx = \frac{1}{8}(4x-1)^{\frac{3}{2}} - \frac{1}{8}(4x-1)^{\frac{1}{2}} + C$$

$$8. \int \frac{1-2x}{1+3x} dx = \frac{5}{9}\ln|1+3x| - \frac{2}{9}(1+3x) + C$$

$$9. \int \frac{6x^{\frac{1}{2}}}{2x^{\frac{3}{2}}+3} dx = 2\ln|2x^{\frac{3}{2}}+3| + C$$

$$10. \int \frac{x^{\frac{3}{2}}}{\sqrt{1-3x^{\frac{5}{2}}}} dx = -\frac{4}{15}\sqrt{1-3x^{\frac{5}{2}}} + C$$

1. $\int (12x(5x-3)^3) dx = \int (24x^2) u^3 \frac{du}{dx} = \int \frac{24}{5} u^3 du = \frac{24}{5} \cdot \frac{1}{4} u^4 + C = \frac{6}{5} (5x-3)^4 + C$

2. $\int \frac{12x}{2x-1} dx = \int \frac{6(2x-1) + 6}{2x-1} dx = \int \frac{6(2x-1)}{2x-1} dx + \int \frac{6}{2x-1} dx = 6x + 3 \ln|2x-1| + C$

3. $\int 2(x^2-1)^5 dx = \int 2u^5 \frac{du}{dx} = \int \frac{2}{5} u^5 du = \frac{2}{5} \cdot \frac{1}{6} u^6 + C = \frac{1}{15} (x^2-1)^6 + C$

4. $\int \frac{5x^2}{2x^2+7} dx = \int \frac{5}{2} \frac{x^2}{x^2 + \frac{7}{2}} dx = \int \frac{5}{2} \left(1 - \frac{7/2}{x^2 + 7/2} \right) dx = \frac{5}{2} x - \frac{5}{2} \int \frac{1}{x^2 + 7/2} dx = \frac{5}{2} x - \frac{5}{2} \cdot \frac{1}{\sqrt{7/2}} \arctan\left(\frac{x\sqrt{2}}{\sqrt{7}}\right) + C$

5. $\int 2x(1-5x)^3 dx = \int 2x u^3 \frac{du}{dx} = -\frac{2}{5} \int u^3 du = -\frac{2}{5} \cdot \frac{1}{4} u^4 + C = -\frac{1}{10} (1-5x)^4 + C$

6. $\int \frac{9x}{\sqrt{4x^2+1}} dx = \int \frac{9}{2} \frac{2x}{\sqrt{4x^2+1}} dx = \frac{9}{2} \int \frac{du}{u} = \frac{9}{2} \ln|u| + C = \frac{9}{2} \ln|2x^2+1| + C$

7. $\int \frac{3x-1}{\sqrt{4x-1}} dx = \int \frac{3u-1}{u^2} \frac{du}{4} = \frac{1}{4} \int \left(\frac{3}{u} - \frac{1}{u^2} \right) du = \frac{3}{4} \ln|u| + \frac{1}{4} u + C = \frac{3}{4} \ln|4x-1| + \frac{1}{4} (4x-1) + C = \frac{3}{4} \ln|4x-1| + x - \frac{1}{4} + C$

8. $\int \frac{1-2x}{1+3x} dx = \int \frac{1-2x}{u} \frac{du}{3} = \frac{1}{3} \int \left(\frac{1}{u} - 2 \right) du = \frac{1}{3} \ln|u| - \frac{2}{3} u + C = \frac{1}{3} \ln|1+3x| - \frac{2}{3} (1+3x) + C = \frac{1}{3} \ln|1+3x| - 2x - \frac{2}{3} + C$

9. $\int \frac{6x^2}{2x^3+3} dx = \int \frac{2}{u} \frac{du}{2} = \int \frac{1}{u} du = \ln|u| + C = \ln|2x^3+3| + C$

10. $\int \frac{2x^2}{\sqrt{1-3x^2}} dx = \int \frac{2}{\sqrt{1-3u^2}} \frac{du}{2} = \int \frac{1}{\sqrt{1-3u^2}} du = \frac{1}{\sqrt{3}} \arcsin(\sqrt{3}u) + C = \frac{1}{\sqrt{3}} \arcsin(\sqrt{3}x) + C$

Question 4

Carry out the following integrations to the answers given, by using substitution only.

$$1. \int_0^{\frac{1}{2}} 8x(2x-1)^4 dx = \frac{1}{15}$$

$$2. \int_2^3 \frac{3x}{3x-5} dx = 1 + \frac{10}{3} \ln 2$$

$$3. \int_0^1 x(1-x^2)^{\frac{3}{2}} dx = \frac{1}{5}$$

$$4. \int_0^1 \frac{4x}{x^2+1} dx = 2 \ln 2$$

$$5. \int_1^3 2x(3x-1)^4 dx = \frac{55808}{5}$$

$$6. \int_4^8 \frac{6x}{\sqrt{2x-7}} dx = 68$$

$$7. \int_0^1 \frac{x}{\sqrt{9-5x^2}} dx = \frac{1}{5}$$

$$8. \int_0^3 \frac{5-2x}{x+1} dx = 14 \ln 2 - 6$$

$$9. \int_0^{\frac{1}{5}} \frac{10x^2}{5x+1} dx = \frac{\ln 4 - 1}{25}$$

$$10. \int_{-\frac{3}{2}}^{-\frac{1}{2}} \frac{5x-2}{2x-5} dx = \frac{5}{2} + \frac{21}{4} \ln\left(\frac{3}{4}\right)$$

1. $\int_0^1 8x(2x-1)^4 dx = \int_0^1 8x u^4 \frac{du}{dx} = \int_0^1 4x u^4 du$
 $= \int_0^1 (2u+1)u^4 du = \int_0^1 2u^5 + u^4 du = \left[\frac{2}{6}u^6 + \frac{1}{5}u^5 \right]_0^1$
 $= \left[\frac{1}{3} + \frac{1}{5} \right] = \frac{8}{15}$

2. $\int_0^3 \frac{2x}{2x-5} dx = \int_0^3 \frac{2x}{u} \frac{du}{dx} = \int_0^3 \frac{u+5}{u} du = \int_0^3 \left(1 + \frac{5}{u} \right) du$
 $= \left[u + 5 \ln|u| \right]_0^3 = \left[3 + 5 \ln 3 \right] - \left[0 + 5 \ln 1 \right] = 3 + 5 \ln 3$

3. $\int_0^1 x(1-x^2)^3 dx = \int_0^1 x u^3 \frac{du}{dx} = \int_0^1 \frac{1}{2} u^3 du$
 $= \left[\frac{1}{2} \cdot \frac{1}{4} u^4 \right]_0^1 = \frac{1}{8}$

4. $\int_0^1 \frac{dx}{x^2+1} = \int_0^1 \frac{1}{u} \frac{du}{dx} = \int_0^1 \frac{1}{u} du$
 $= \left[\ln|u| \right]_0^1 = \ln 2 - \ln 1 = \ln 2$

5. $\int_1^3 2x(3x-1)^5 dx = \int_1^3 2x u^5 \frac{du}{dx} = \int_1^3 \frac{2}{3} u^5 du$
 $= \left[\frac{2}{3} \cdot \frac{1}{6} u^6 \right]_1^3 = \left[\frac{1}{9} u^6 \right]_1^3 = \frac{1}{9} (3^6 - 1^6) = \frac{728}{9}$

6. $\int_1^4 \frac{dx}{\sqrt{2x-7}} = \int_1^4 \frac{1}{u} \frac{du}{dx} = \int_1^4 \frac{1}{u} du$
 $= \left[\ln|u| \right]_1^4 = \ln 4 - \ln 1 = \ln 4$

7. $\int_0^1 \frac{x}{\sqrt{9-x^2}} dx = \int_0^1 \frac{x}{u} \frac{du}{dx} = \int_0^1 \frac{1}{2} \frac{1}{u} du$
 $= \frac{1}{2} \left[\ln|u| \right]_0^1 = \frac{1}{2} \ln 2$

8. $\int_0^3 \frac{5-2x}{x+1} dx = \int_0^3 \frac{5-2x}{u} \frac{du}{dx} = \int_0^3 \frac{5-2(u-1)}{u} du$
 $= \int_0^3 \frac{7-2u}{u} du = \int_0^3 \left(\frac{7}{u} - 2 \right) du = \left[7 \ln|u| - 2u \right]_0^3$
 $= 7 \ln 3 - 6$

9. $\int_0^1 \frac{10x^2}{2x+1} dx = \int_0^1 \frac{10x^2}{u} \frac{du}{dx} = \int_0^1 \frac{20x^2}{u} dx$
 $= \int_0^1 \frac{20 \left(\frac{u-1}{2} \right)^2}{u} du = \int_0^1 \frac{5(u-1)^2}{u} du$
 $= \int_0^1 \left(5 - \frac{10}{u} + 5 \right) du = \left[10u - 10 \ln|u| \right]_0^1 = 10 - 10 \ln 1 = 10$

10. $\int_0^1 \frac{x-2}{2x-5} dx = \int_0^1 \frac{x-2}{u} \frac{du}{dx} = \int_0^1 \frac{1}{2} \frac{u-4}{u} du$
 $= \frac{1}{2} \int_0^1 \left(1 - \frac{4}{u} \right) du = \frac{1}{2} \left[u - 4 \ln|u| \right]_0^1 = \frac{1}{2} (1 - 4 \ln 1) = \frac{1}{2}$

Question 5

Carry out the following integrations to the answers given, by using substitution only.

$$1. \int_0^{\frac{3}{2}} 2x(2x-3)^4 dx = \frac{243}{20}$$

$$2. \int_0^2 \frac{4x}{4x+1} dx = 2 - \frac{1}{2} \ln 3$$

$$3. \int_0^1 x^2(1-x^3)^{\frac{9}{2}} dx = \frac{2}{33}$$

$$4. \int_0^4 \frac{12x}{x^2+9} dx = 12 \ln\left(\frac{5}{3}\right)$$

$$5. \int_1^2 2x(3x-1)^4 dx = \frac{3569}{5}$$

$$6. \int_2^6 \frac{6x}{\sqrt{3x-2}} dx = \frac{272}{9}$$

$$7. \int_0^1 \frac{x}{\sqrt{16-7x^2}} dx = \frac{1}{7}$$

$$8. \int_5^6 \frac{1-2x}{x-4} dx = -2 - 7 \ln 2$$

$$9. \int_0^{\frac{1}{3}} \frac{9x^2}{3x+1} dx = \frac{\ln 4 - 1}{6}$$

$$10. \int_0^{\frac{3}{2}} \frac{2x-3}{2x+3} dx = \frac{3}{2}(1 - \ln 4)$$

1. $\int_0^2 2x(2x-3)^5 dx = \int_{-3}^7 2x u^5 \frac{du}{dx} = \int_{-3}^7 (u+3)u^5 \frac{1}{2} du$
 $= \frac{1}{2} \int_{-3}^7 u^6 + 3u^5 du = \frac{1}{2} \left[\frac{1}{7} u^7 + \frac{3}{6} u^6 \right]_{-3}^7 = \frac{1}{2} \left[\left(\frac{7^7}{7} + \frac{3 \cdot 7^6}{6} \right) - \left(\frac{(-3)^7}{7} + \frac{3 \cdot (-3)^6}{6} \right) \right]$
 $= \frac{1}{2} \times \frac{253}{20} = \frac{253}{40}$

2. $\int_0^2 \frac{4x}{4x+1} dx = \int_1^9 \frac{\frac{1}{2}u}{\frac{1}{2}u+1} \frac{du}{2} = \frac{1}{4} \int_1^9 \frac{u-1+1}{u+1} du$
 $= \frac{1}{4} \int_1^9 \left(1 - \frac{1}{u+1} \right) du = \frac{1}{4} \left[u - \ln|u+1| \right]_1^9 = \frac{1}{4} \left[(9-1) - (\ln 10 - \ln 2) \right]$
 $= \frac{1}{4} [8 - \ln 5] = 2 - \frac{1}{4} \ln 5$

3. $\int_0^1 x^2(1-2x)^{\frac{5}{2}} dx = \int_1^0 x^2 u^{\frac{5}{2}} \frac{du}{-2} = -\frac{1}{2} \int_1^0 u^{\frac{5}{2}} du$
 $= \frac{1}{2} \left[\frac{2}{7} u^{\frac{7}{2}} \right]_1^0 = \frac{1}{2} \left[0 - \frac{2}{7} \right] = -\frac{1}{7}$

4. $\int_0^8 \frac{12x}{x^2+9} dx = \int_9^{25} \frac{2u}{u} \frac{du}{2} = \int_9^{25} \frac{du}{u} = \ln u \Big|_9^{25}$
 $= \ln 25 - \ln 9 = \ln \left(\frac{25}{9} \right) = 2 \ln \frac{5}{3}$

5. $\int_1^2 2x(3x-1)^4 dx = \int_{-1}^5 2x u^4 \frac{du}{3} = \frac{2}{3} \int_{-1}^5 u^4 du$
 $= \frac{2}{3} \left[\frac{1}{5} u^5 \right]_{-1}^5 = \frac{2}{3} \left[\frac{1}{5} (5^5 - (-1)^5) \right] = \frac{2}{3} \left[\frac{1}{5} (3125 + 1) \right] = \frac{2}{3} \times \frac{3126}{5} = \frac{3504}{5}$

6. $\int_2^6 \frac{6x}{43x-2} dx = \int_2^6 \frac{6x}{4} \frac{du}{du} = \frac{3}{2} \int_2^6 \frac{u}{u^2} du = \frac{3}{2} \int_2^6 \frac{1}{u} du$
 $= \frac{3}{2} \left[\ln u \right]_2^6 = \frac{3}{2} \ln \frac{6}{2} = \frac{3}{2} \ln 3$

7. $\int_0^2 \frac{x}{\sqrt{6-2x}} dx = \int_0^2 \frac{x}{\sqrt{6-2x}} \frac{du}{-2} = -\frac{1}{2} \int_0^2 \frac{u}{\sqrt{6-2x}} du$
 $= -\frac{1}{2} \int_6^2 \frac{u}{\sqrt{u}} du = -\frac{1}{2} \int_6^2 u^{\frac{1}{2}} du = -\frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_6^2 = -\frac{1}{3} \left[2^{\frac{3}{2}} - 6^{\frac{3}{2}} \right] = \frac{1}{3} (6\sqrt{6} - 2\sqrt{2})$

8. $\int_1^2 \frac{1-2x}{2-x} dx = \int_1^2 \frac{1-2(u+1)}{u} du = \int_1^2 \frac{-1-2u}{u} du = \int_1^2 \left(-\frac{1}{u} - 2 \right) du$
 $= \left[-\ln|u| - 2u \right]_1^2 = (-\ln 2 - 4) - (-\ln 1 - 2) = -\ln 2 - 2$

9. $\int_0^2 \frac{3x^2}{3x^2+1} dx = \int_1^5 \frac{u-1}{u} \frac{du}{3} = \frac{1}{3} \int_1^5 \left(1 - \frac{1}{u} \right) du$
 $= \frac{1}{3} \left[u - \ln|u| \right]_1^5 = \frac{1}{3} \left[(5-1) - (\ln 5 - \ln 1) \right] = \frac{4}{3} - \frac{1}{3} \ln 5$

10. $\int_0^2 \frac{2x-3}{x^2+3} dx = \int_3^7 \frac{u-3}{u} \frac{du}{2} = \frac{1}{2} \int_3^7 \left(1 - \frac{3}{u} \right) du$
 $= \frac{1}{2} \left[u - 3 \ln|u| \right]_3^7 = \frac{1}{2} \left[(7-3) - (3 \ln 7 - 3 \ln 3) \right] = \frac{1}{2} \left[4 - 3 \ln \frac{7}{3} \right]$

Question 6

Carry out the following integrations.

$$1. \int \frac{x}{\sqrt{x+1}} dx = \frac{2}{3}(x+1)^{\frac{3}{2}} - 2(x+1)^{\frac{1}{2}} + C = \frac{2}{3}(x-2)\sqrt{x+1} + C$$

$$2. \int \frac{2x}{(2x+1)^3} dx = -\frac{1}{2}(2x+1)^{-1} + \frac{1}{4}(2x+1)^{-2} + C = -\frac{4x+1}{4(2x+1)^2} + C$$

$$3. \int \frac{x}{x+1} dx = x - \ln|x+1| + C = x+1 - \ln|x+1| + C$$

$$4. \int \frac{x}{\sqrt{x-1}} dx = \frac{2}{3}(x-1)^{\frac{3}{2}} + 2(x-1)^{\frac{1}{2}} + C = \frac{2}{3}(x+2)\sqrt{x-1} + C$$

$$5. \int \frac{4x+1}{2x-5} dx = 2x + \frac{11}{2} \ln|2x-5| + C$$

$$6. \int \frac{x^2}{2x-1} dx = \frac{1}{16}(2x-1)^2 + \frac{1}{4}(2x-1) + \frac{1}{8} \ln|2x-1| + C = \frac{1}{4}x^2 + \frac{1}{4}x + \frac{1}{8} \ln|2x-1| + C$$

$$7. \int \frac{2x+1}{2x-1} dx = \frac{1}{2}(2x-1) + \ln|2x-1| + C = x + \ln|2x-1| + C$$

$$8. \int \frac{6x}{\sqrt{2x+3}} dx = (2x+3)^{\frac{3}{2}} - 9(2x+3)^{\frac{1}{2}} + C = 2(x-3)\sqrt{2x+3} + C$$

$$9. \int \frac{3x-1}{2x+3} dx = \frac{3}{4}(2x-3) - \frac{11}{4} \ln|2x+3| + C = \frac{3}{2}x - \frac{11}{4} \ln|2x+3| + C$$

$$10. \int \frac{8x^2}{1-2x} dx = -\frac{1}{2}(1-2x)^2 + 2(1-2x) - \ln|1-2x| + C = -2x^2 - 2x - \ln|1-2x| + C$$

1. $\int \frac{x}{\sqrt{x+1}} dx = \int \frac{u-1}{\sqrt{u}} du = \int \frac{u-1}{u^{1/2}} du$
 $= \int u^{1/2} - u^{-1/2} du = \frac{2}{3} u^{3/2} - 2u^{1/2} + C$
 $= \frac{2}{3} (x+1)^{3/2} - 2(x+1)^{1/2} + C$
u = x+1
 $\frac{du}{dx} = 1$
 $dx = du$

2. $\int \frac{2x}{(2x+1)^2} dx = \int \frac{2x}{u^2} \cdot \frac{du}{2} = \int \frac{u-1}{u^2} du$
 $= \int \frac{u}{u^2} - \frac{1}{u^2} du = \int u^{-1} - u^{-2} du = \frac{1}{2} (-u^{-1} + u^{-2})$
 $= \frac{1}{2} \left[\frac{1}{2u} - \frac{1}{u} \right] + C = \frac{1}{4(2x+1)} - \frac{1}{2(2x+1)} + C$
u = 2x+1
 $\frac{du}{dx} = 2$
 $dx = \frac{du}{2}$
 $2x = u-1$

3. $\int \frac{2x}{\sqrt{x+1}} dx = \int \frac{u-1}{\sqrt{u}} du = \int \frac{u-1}{u^{1/2}} du = \int u^{1/2} - u^{-1/2} du$
 $= u - \ln|u| + C = (x+1) - \ln|x+1| + C$
u = x+1
 $\frac{du}{dx} = 1$
 $dx = du$
 $x = u-1$

4. $\int \frac{2x}{\sqrt{x+1}} dx = \int \frac{u-1}{\sqrt{u}} du = \int \frac{u-1}{u^{1/2}} du = \int u^{1/2} - u^{-1/2} du$
 $= \frac{2}{3} u^{3/2} - 2u^{1/2} + C = \frac{2}{3} (x+1)^{3/2} - 2(x+1)^{1/2} + C$
u = x+1
 $\frac{du}{dx} = 1$
 $dx = du$
 $x = u-1$

5. $\int \frac{4x+1}{2x-5} dx = \int \frac{2(2x-5)+11}{2x-5} dx = \int \frac{2(2x-5)}{2x-5} + \frac{11}{2x-5} dx$
 $= \int 2 + \frac{11}{2x-5} dx = \frac{1}{2} \int \frac{11}{u} du$
 $= \frac{1}{2} [2u + 11 \ln|u|] + C = u + \frac{11}{2} \ln|u| + C$
 $= (2x-5) + \frac{11}{2} \ln|2x-5| + C$
u = 2x-5
 $\frac{du}{dx} = 2$
 $dx = \frac{du}{2}$
 $2x = u+5$
 $4x+1 = 2u+11$

6. $\int \frac{x^2}{2x-1} dx = \int \frac{u^2}{u-1} du = \frac{1}{2} \int \frac{(u+1)^2}{u-1} du$
 $= \frac{1}{2} \int \frac{u^2 + 2u + 1}{u-1} du = \frac{1}{2} \int \left(\frac{u^2}{u-1} + \frac{2u}{u-1} + \frac{1}{u-1} \right) du$
 $= \frac{1}{2} \left[\frac{1}{2} u^2 + 2u + \ln|u| \right] + C$
 $= \frac{1}{4} u^2 + u + \ln|u| + C = \frac{1}{4} (2x-1)^2 + (2x-1) + \ln|2x-1| + C$
u = 2x-1
 $\frac{du}{dx} = 2$
 $dx = \frac{du}{2}$
 $2x = u+1$

7. $\int \frac{2x+1}{2x-1} dx = \int \frac{u+2}{u-1} du = \frac{1}{2} \int \frac{u+2}{u-1} du$
 $= \frac{1}{2} \int \left(\frac{u}{u-1} + \frac{2}{u-1} \right) du = \frac{1}{2} \left[\frac{1}{2} u^2 + 2 \ln|u| \right] + C$
 $= \frac{1}{4} u^2 + \ln|u| + C = \frac{1}{4} (2x-1)^2 + \ln|2x-1| + C$
u = 2x-1
 $\frac{du}{dx} = 2$
 $dx = \frac{du}{2}$
 $2x = u+1$

8. $\int \frac{6x}{\sqrt{2x+3}} dx = \int \frac{3u-9}{u} du = \frac{1}{2} \int \frac{3u-9}{u} du$
 $= \frac{1}{2} \int \left(3 - \frac{9}{u} \right) du = \frac{1}{2} \left[3u - 9 \ln|u| \right] + C$
 $= \frac{3}{2} u - \frac{9}{2} \ln|u| + C = \frac{3}{2} (2x+3) - \frac{9}{2} \ln|2x+3| + C$
u = 2x+3
 $\frac{du}{dx} = 2$
 $dx = \frac{du}{2}$
 $2x = u-3$
 $6x = 3u-9$

9. $\int \frac{2x-1}{\sqrt{x+3}} dx = \int \frac{u-4}{u} du = \frac{1}{2} \int \frac{u-4}{u} du$
 $= \frac{1}{2} \int \left(1 - \frac{4}{u} \right) du = \frac{1}{2} \left[u - 4 \ln|u| \right] + C$
 $= \frac{1}{2} u - 2 \ln|u| + C = \frac{1}{2} (2x+3) - 2 \ln|2x+3| + C$
u = 2x+3
 $\frac{du}{dx} = 2$
 $dx = \frac{du}{2}$
 $2x = u-3$
 $2x = u-3$

10. $\int \frac{5x^2}{(x-2)^2} dx = \int \frac{5u^2}{u^2} \cdot \frac{du}{u} = \frac{1}{2} \int \frac{5u^2}{u} du$
 $= \frac{1}{2} \int \left(5 - \frac{20}{u} \right) du = \frac{1}{2} \left[5u - 20 \ln|u| \right] + C$
 $= \frac{5}{2} u - 10 \ln|u| + C = \frac{5}{2} (x-2) - 10 \ln|x-2| + C$
u = x-2
 $\frac{du}{dx} = 1$
 $dx = du$
 $x = u+2$
 $5x = 5u+10$
 $4x+1 = 2u+11$

Question 7

Carry out the following integrations **using the substitutions** given.

1. $\int x\sqrt{1-x} \, dx = \frac{2}{5}(1-x)^{\frac{5}{2}} - \frac{2}{3}(1-x)^{\frac{3}{2}} + C$ Use $u = 1-x$, or $u = \sqrt{1-x}$

2. $\int \frac{6x}{\sqrt{2x+1}} \, dx = (2x+1)^{\frac{3}{2}} - 3(2x+1)^{\frac{1}{2}} + C$ Use $u = 2x+1$, or $u = \sqrt{2x+1}$

3. $\int \cos^3 x \, dx = \sin x - \frac{1}{3}\sin^3 x + C$ Use $u = \sin x$

4. $\int \sec^4 x \, dx = \tan x + \frac{1}{3}\tan^3 x + C$ Use $u = \tan x$

5. $\int \frac{1}{\sqrt{x}(x-4)} \, dx = \frac{1}{2} \ln \left| \frac{\sqrt{x}-2}{\sqrt{x}+2} \right| + C$ Use $u = \sqrt{x}$

6. $\int \frac{\sqrt{x^2+9}}{x} \, dx = \sqrt{x^2+9} + \frac{3}{2} \ln \left| \frac{\sqrt{x^2+9}-3}{\sqrt{x^2+9}+3} \right| + C$ Use $u = \sqrt{x^2+9}$

7. $\int \frac{1+\cos x}{\sin x} \, dx = \ln |\cos x - 1| + C$ Use $u = \cos x$

8. $\int \frac{1}{1+\sqrt{x-2}} \, dx = 2\sqrt{x-2} + 2\ln(1+\sqrt{x-2}) + C$ Use $u = \sqrt{x-2}$

9. $\int \sec^2 x \tan x \sqrt{1+\tan x} \, dx = \frac{2}{15}(3\tan x - 2)(1+\tan x)^{\frac{3}{2}} + C$ Use $u = \sqrt{1+\tan x}$

10. $\int \frac{9}{\sqrt{x}(9x-1)} \, dx = 3 \ln \left| \frac{3\sqrt{x}-1}{3\sqrt{x}+1} \right| + C$ Use $u = \sqrt{x}$

$x = \tan \theta$

$$10. \int \frac{9}{\sqrt{x}(\sqrt{x}-1)} dx = \int \frac{9}{\sqrt{x}(\sqrt{x}-1)} \cdot 2(\sqrt{x}-1) d(\sqrt{x}-1) = \int \frac{18}{\sqrt{x}-1} d(\sqrt{x}-1)$$

$$= \int \frac{18}{3(\sqrt{x}-1)^3} d(\sqrt{x}-1)$$

$$\begin{aligned} & \text{BY METHOD OF PARTIALS} \\ & \frac{18}{3(\sqrt{x}-1)^3} = \frac{A}{\sqrt{x}-1} + \frac{B}{(\sqrt{x}-1)^2} + \frac{C}{(\sqrt{x}-1)^3} \\ & \text{Equate } (1) \text{ and } (2) \Rightarrow \end{aligned}$$

$$\begin{aligned} & 6 = \frac{A}{\sqrt{x}-1} + \frac{B}{(\sqrt{x}-1)^2} + \frac{C}{(\sqrt{x}-1)^3} \\ & \Rightarrow 6(\sqrt{x}-1)^3 = A(\sqrt{x}-1)^2 + B(\sqrt{x}-1) + C \end{aligned}$$

$$\begin{aligned} & 6 = \frac{A}{\sqrt{x}-1} + \frac{B}{(\sqrt{x}-1)^2} + \frac{C}{(\sqrt{x}-1)^3} \\ & \Rightarrow 6(\sqrt{x}-1)^3 = A(\sqrt{x}-1)^2 + B(\sqrt{x}-1) + C \end{aligned}$$

$$= \int \frac{9}{3(\sqrt{x}-1)^3} d(\sqrt{x}-1) = 3 \ln|\sqrt{x}-1| - 3 \ln|\sqrt{x}-1| = 3 \ln \left| \frac{\sqrt{x}-1}{\sqrt{x}-1} \right| + C$$

$$= 3 \ln \left| \frac{\sqrt{x}-1}{\sqrt{x}-1} \right| + C$$

[illegible]

Question 8

Carry out the following integrations.

$$1. \int \frac{1}{2+\sqrt{x}} dx = 2\sqrt{x} - 4 \ln|2+\sqrt{x}| + C$$

$$2. \int \frac{x^2}{1-2x} dx = -\frac{1}{4}x(x+1) - \frac{1}{8} \ln|1-2x| = -\frac{1}{16}(1-2x)^2 + \frac{1}{4}(1-2x) - \frac{1}{8} \ln|1-2x| + C$$

$$3. \int \frac{3x^2+2}{4x+1} dx = \frac{1}{64}(24x^2-12x+35 \ln|4x+1|) + C = \frac{3}{128}(4x+1)^2 - \frac{3}{32}(4x+1) + \frac{35}{64} \ln|4x+1|$$

$$4. \int \frac{4-3x}{2x+1} dx = -\frac{3}{2}x + \frac{11}{4} \ln|2x+1| + C$$

$$5. \int \frac{x+1}{x-5} dx = x + 6 \ln|x-5| + C$$

$$6. \int \frac{x^2}{x-2} dx = \frac{1}{2}x^2 + 2x + 4 \ln|x-2| + C$$

$$7. \int \frac{1}{2+\sqrt{x-1}} dx = 2\sqrt{x-1} - 4 \ln|\sqrt{x-1}+2| + C$$

$$8. \int \frac{x+4}{x-4} dx = x + 8 \ln|x-4| + C$$

Question 9

Carry out the following integrations.

$$1. \int x\sqrt{x+1} \, dx = \frac{2}{15}(3x+2)\sqrt{(x+1)^3} + C$$

$$2. \int \frac{x+1}{\sqrt[3]{x^2+2x+3}} \, dx = \frac{3}{4}\sqrt[3]{(x^2+2x+3)^2} + C$$

$$3. \int \frac{3x^3+5x}{x^2+1} \, dx = \frac{3}{2}x^2 + \ln|x^2+1| + C$$

$$4. \int \frac{2x+1}{3x-1} \, dx = \frac{2}{3}x + \frac{5}{9}\ln|3x-1| + C$$

$$5. \int \frac{1}{(x-1)\sqrt{x^2-1}} \, dx = -\sqrt{\frac{x+1}{x-1}} + C, \text{ use } x-1 = \frac{1}{u}$$

$$6. \int \frac{4x^3\sqrt{x^4+1}}{1+\sqrt{x^4+1}} \, dx = x^4 - 2\sqrt{x^4+1} + 2\ln|1+\sqrt{x^4+1}| + C$$

Question 10

Carry out the following integrations to the answers given.

$$1. \int_0^{\frac{1}{2}} \frac{x}{(2-x)^2} dx = \frac{1}{3} + \ln\left(\frac{3}{4}\right)$$

$$2. \int_1^2 \frac{x}{(2x-1)^2} dx = \frac{2 + \ln 27}{12}$$

$$3. \int_0^2 \frac{x+2}{\sqrt{4x+1}} dx = \frac{17}{6}$$

$$4. \int_0^{36} \frac{1}{\sqrt{x}(\sqrt{x}+2)} dx = \ln 16$$

$$5. \int_{-6}^{\frac{3}{2}} \frac{x}{\sqrt{4-2x}} dx = -\frac{9}{2}$$

$$6. \int_1^5 \frac{x+1}{(2x-1)^{\frac{3}{2}}} dx = 2$$

$$7. \int_{\frac{2}{3}}^1 \frac{x}{2x-1} dx = \frac{1}{6} + \frac{1}{4} \ln 3$$

$$8. \int_{-1}^7 \frac{x^2}{\sqrt{x+2}} dx = \frac{652}{15}$$

$$9. \int_1^{\frac{5}{2}} \frac{4x}{\sqrt{2x-1}} dx = \frac{20}{3}$$

$$10. \int_0^1 \frac{x}{(1+x)^2} dx = -\frac{1}{2} + \ln 2$$

Question 11

Carry out the following integrations to the answers given.

$$1. \int_0^3 x\sqrt{x+1} \, dx = \frac{116}{15}$$

$$2. \int_0^2 \frac{6x^3}{\sqrt{x^2+1}} \, dx = 4(1+\sqrt{5})$$

$$3. \int_{-1}^0 \frac{x^2}{1-x} \, dx = -\frac{1}{2} + \ln 2$$

$$4. \int_0^{100} \frac{1}{20-\sqrt{x}} \, dx = 40\ln 2 - 20$$

$$5. \int_0^{\frac{1}{4}} 2x\sqrt{1-4x} \, dx = \frac{1}{30}$$

$$6. \int_0^{\frac{\pi}{2}} \sin x \cos x (1+\sin x)^5 \, dx = \frac{107}{14}$$

$$7. \int_2^5 \frac{x^2}{\sqrt{x-1}} \, dx = \frac{356}{15}$$

Question 12

Carry out the following integrations to the answers given.

1. $\int_0^{\sqrt{2}} \frac{x^2}{\sqrt{4-x^2}} dx = \frac{\pi}{2} - 1$, use $x = 2 \sin \theta$

2. $\int_1^{\sqrt{2}} \frac{1}{x^2 \sqrt{4-x^2}} dx = \frac{1}{4}(\sqrt{3}-1)$, use $x = 2 \cos \theta$

3. $\int_0^1 \frac{1}{(1+x^2)^2} dx = \frac{1}{8}(\pi+2)$, use $x = \tan \theta$

4. $\int_{\sqrt{2}}^2 \frac{1}{x^2 \sqrt{x^2 - 1}} dx = \frac{1}{2}(\sqrt{3} - \sqrt{2}), \text{ use } x = \sec \theta$

Handwritten solution for problem 4:

$$\int_{\sqrt{2}}^2 \frac{1}{x^2 \sqrt{x^2 - 1}} dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{\sec^2 \theta \sqrt{\sec^2 \theta - 1}} (\sec \theta) d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec \theta \cos \theta}{\sec \theta \sqrt{\tan^2 \theta}} d\theta = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{\sec \theta \tan \theta} d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{\sec \theta} d\theta = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \cos \theta d\theta = \left[\sin \theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$$

$$= \sin \frac{\pi}{3} - \sin \frac{\pi}{4} = \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} = \frac{1}{2}(\sqrt{3} - \sqrt{2})$$

Side notes:

- $x = \sec \theta$
- $\frac{dx}{d\theta} = \sec \theta \tan \theta$
- $dx = \sec \theta \tan \theta d\theta$
- $x = 2 \Rightarrow \sec \theta = 2 \Rightarrow \theta = \frac{\pi}{3}$
- $x = \sqrt{2} \Rightarrow \sec \theta = \sqrt{2} \Rightarrow \theta = \frac{\pi}{4}$

5. $\int_0^{\frac{3}{4}} \frac{1}{\sqrt{3-4x^2}} dx = \frac{\pi}{6}, \text{ use } x = \frac{\sqrt{3}}{2} \sin \theta$

Handwritten solution for problem 5:

$$\int_0^{\frac{3}{4}} \frac{1}{\sqrt{3-4x^2}} dx = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{3-4\left(\frac{\sqrt{3}}{2} \sin \theta\right)^2}} \left(\frac{\sqrt{3}}{2} \cos \theta\right) d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{\frac{\sqrt{3}}{2} \cos \theta}{\sqrt{3-3 \sin^2 \theta}} d\theta = \int_0^{\frac{\pi}{2}} \frac{\frac{\sqrt{3}}{2} \cos \theta}{\sqrt{3(1-\sin^2 \theta)}} d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{\frac{\sqrt{3}}{2} \cos \theta}{\sqrt{3} \cos \theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{1}{2} d\theta = \left[\frac{1}{2} \theta \right]_0^{\frac{\pi}{2}}$$

$$= \left(\frac{1}{2} \times \frac{\pi}{2} \right) - (0) = \frac{\pi}{4}$$

Side notes:

- $x = \frac{\sqrt{3}}{2} \sin \theta$
- $\frac{dx}{d\theta} = \frac{\sqrt{3}}{2} \cos \theta$
- $dx = \frac{\sqrt{3}}{2} \cos \theta d\theta$
- $x = 0 \Rightarrow \sin \theta = 0 \Rightarrow \theta = 0$
- $x = \frac{3}{4} \Rightarrow \frac{\sqrt{3}}{2} \sin \theta = \frac{3}{4} \Rightarrow \sin \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{2}$

6. $\int_0^1 \frac{1}{(1+3x^2)^{\frac{3}{2}}} dx = \frac{1}{2}, \text{ use } x = \frac{1}{\sqrt{3}} \tan \theta$

Handwritten solution for problem 6:

$$\int_0^1 \frac{1}{(1+3x^2)^{\frac{3}{2}}} dx = \int_0^{\frac{\pi}{4}} \frac{1}{\left(1+3\left(\frac{1}{\sqrt{3}} \tan \theta\right)^2\right)^{\frac{3}{2}}} \left(\frac{1}{\sqrt{3}} \sec^2 \theta\right) d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{\frac{1}{\sqrt{3}} \sec^2 \theta}{\left(1+\tan^2 \theta\right)^{\frac{3}{2}}} d\theta = \int_0^{\frac{\pi}{4}} \frac{\frac{1}{\sqrt{3}} \sec^2 \theta}{\sec^3 \theta} d\theta = \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt{3} \sec \theta} d\theta$$

$$= \frac{1}{\sqrt{3}} \int_0^{\frac{\pi}{4}} \cos \theta d\theta = \frac{1}{\sqrt{3}} \left[\sin \theta \right]_0^{\frac{\pi}{4}} = \frac{1}{\sqrt{3}} \left(\sin \frac{\pi}{4} - 0 \right) = \frac{1}{\sqrt{3}} \times \frac{\sqrt{2}}{2} = \frac{1}{2}$$

Side notes:

- $x = \frac{1}{\sqrt{3}} \tan \theta$
- $\frac{dx}{d\theta} = \frac{1}{\sqrt{3}} \sec^2 \theta$
- $dx = \frac{1}{\sqrt{3}} \sec^2 \theta d\theta$
- $x = 0 \Rightarrow \tan \theta = 0 \Rightarrow \theta = 0$
- $x = 1 \Rightarrow \frac{1}{\sqrt{3}} \tan \theta = 1 \Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{4}$

7. $\int_0^1 \frac{1}{\sqrt{2-x^2}} dx = \frac{\pi}{4}$, use $x = \sqrt{2} \sin \theta$

Handwritten solution for problem 7:

$$\int_0^1 \frac{1}{\sqrt{2-x^2}} dx = \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt{2-(\sqrt{2}\sin\theta)^2}} \sqrt{2}\cos\theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sqrt{2}\cos\theta}{\sqrt{2-2\sin^2\theta}} d\theta = \int_0^{\frac{\pi}{4}} \frac{\sqrt{2}\cos\theta}{\sqrt{2}\cos\theta} d\theta = \int_0^{\frac{\pi}{4}} 1 d\theta = \left[\theta\right]_0^{\frac{\pi}{4}} = \frac{\pi}{4}$$

Substitution details:

$$x = \sqrt{2}\sin\theta$$

$$\frac{dx}{d\theta} = \sqrt{2}\cos\theta$$

$$x=1 \Rightarrow \sin\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}$$

$$x=0 \Rightarrow \sin\theta = 0 \Rightarrow \theta = 0$$

8. $\int_0^{\frac{1}{2}} \frac{1}{4x^2+3} dx = \frac{\pi\sqrt{3}}{36}$, use $x = \frac{\sqrt{3}}{2} \tan \theta$

Handwritten solution for problem 8:

$$\int_0^{\frac{1}{2}} \frac{1}{4x^2+3} dx = \int_0^{\frac{\pi}{6}} \frac{1}{4\left(\frac{\sqrt{3}}{2}\tan\theta\right)^2+3} \frac{\sqrt{3}}{2} \sec^2\theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{\frac{\sqrt{3}}{2} \sec^2\theta}{4\left(\frac{3}{4}\tan^2\theta\right)+3} d\theta = \int_0^{\frac{\pi}{6}} \frac{\frac{\sqrt{3}}{2} \sec^2\theta}{3(1+\tan^2\theta)} d\theta = \int_0^{\frac{\pi}{6}} \frac{\frac{\sqrt{3}}{2} \sec^2\theta}{3\sec^2\theta} d\theta = \int_0^{\frac{\pi}{6}} \frac{\sqrt{3}}{6} d\theta = \left[\frac{\sqrt{3}}{6}\theta\right]_0^{\frac{\pi}{6}} = \frac{\sqrt{3}}{6} \cdot \frac{\pi}{6} = \frac{\pi\sqrt{3}}{36}$$

Substitution details:

$$x = \frac{\sqrt{3}}{2}\tan\theta$$

$$\frac{dx}{d\theta} = \frac{\sqrt{3}}{2}\sec^2\theta$$

$$x=\frac{1}{2} \Rightarrow \tan\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = \frac{\pi}{6}$$

$$x=0 \Rightarrow \tan\theta = 0 \Rightarrow \theta = 0$$

9. $\int_0^1 \frac{1}{(4-x^2)^{\frac{3}{2}}} dx = \frac{\sqrt{3}}{12}$, use $x = 2 \sin \theta$

Handwritten solution for problem 9:

$$\int_0^1 \frac{1}{(4-x^2)^{\frac{3}{2}}} dx = \int_0^{\frac{\pi}{6}} \frac{1}{(4-(2\sin\theta)^2)^{\frac{3}{2}}} (2\cos\theta) d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{2\cos\theta}{(4(1-\sin^2\theta))^{\frac{3}{2}}} d\theta = \int_0^{\frac{\pi}{6}} \frac{2\cos\theta}{(4\cos^2\theta)^{\frac{3}{2}}} d\theta = \int_0^{\frac{\pi}{6}} \frac{2\cos\theta}{8\cos^3\theta} d\theta = \int_0^{\frac{\pi}{6}} \frac{1}{4\cos^2\theta} d\theta = \left[\frac{1}{4}\tan\theta\right]_0^{\frac{\pi}{6}} = \frac{1}{4}\tan\frac{\pi}{6} = \frac{1}{4} \cdot \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{12}$$

Substitution details:

$$x = 2\sin\theta$$

$$\frac{dx}{d\theta} = 2\cos\theta$$

$$x=1 \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$$

$$x=0 \Rightarrow \sin\theta = 0 \Rightarrow \theta = 0$$

10. $\int_{\sqrt{2}}^2 \frac{\sqrt{x^2-1}}{x} dx = \sqrt{3} - 1 - \frac{\pi}{12}$, use $x = \operatorname{cosec} \theta$

Handwritten solution for problem 10:

$$\int_{\sqrt{2}}^2 \frac{\sqrt{x^2-1}}{x} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sqrt{\operatorname{cosec}^2 \theta - 1}}{\operatorname{cosec} \theta} \operatorname{cosec} \theta d\theta$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sqrt{\tan^2 \theta} d\theta = \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \tan \theta d\theta = \left[-\ln |\cos \theta| \right]_{\frac{\pi}{6}}^{\frac{\pi}{4}} = \left(-\ln \frac{1}{\sqrt{2}} \right) - \left(-\ln \frac{\sqrt{3}}{2} \right)$$

$$= \ln \frac{2}{\sqrt{2}} - \ln \frac{\sqrt{3}}{2} = \ln \sqrt{2} - \ln \frac{\sqrt{3}}{2} = \ln \sqrt{2} - \ln \sqrt{3} + \ln 2 = \ln 2 - \ln \sqrt{3} + \ln \sqrt{2} = \ln 2 - \ln \sqrt{3} + \frac{1}{2} \ln 2 = \frac{3}{2} \ln 2 - \ln \sqrt{3} = \ln 2 - \frac{1}{2} \ln 3 = \ln 2 - \frac{1}{2} \ln 3$$

Substitution: $x = \operatorname{cosec} \theta$
 $\frac{dx}{d\theta} = -\operatorname{cosec} \theta \cot \theta$
 $dx = -\operatorname{cosec} \theta \cot \theta d\theta$
 $x = 2 \Rightarrow \theta = \frac{\pi}{6}$
 $x = \sqrt{2} \Rightarrow \theta = \frac{\pi}{4}$

11. $\int_0^1 \frac{1}{\sqrt{4-3x^2}} dx = \frac{\pi\sqrt{3}}{9}$, use $x = \frac{2}{\sqrt{3}} \sin \theta$

Handwritten solution for problem 11:

$$\int_0^1 \frac{1}{\sqrt{4-3x^2}} dx = \int_0^{\frac{\pi}{6}} \frac{1}{\sqrt{4-3\left(\frac{2}{\sqrt{3}} \sin \theta\right)^2}} \times \frac{2}{\sqrt{3}} \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{\sqrt{4-4\sin^2 \theta}} \times \frac{2}{\sqrt{3}} \cos \theta d\theta = \int_0^{\frac{\pi}{6}} \frac{1}{\sqrt{4\cos^2 \theta}} \times \frac{2}{\sqrt{3}} \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{2\cos \theta} \times \frac{2}{\sqrt{3}} \cos \theta d\theta = \int_0^{\frac{\pi}{6}} \frac{1}{\sqrt{3}} d\theta = \left[\frac{\theta}{\sqrt{3}} \right]_0^{\frac{\pi}{6}} = \frac{\frac{\pi}{6}}{\sqrt{3}} - 0 = \frac{\pi}{6\sqrt{3}} = \frac{\pi\sqrt{3}}{18}$$

Substitution: $x = \frac{2}{\sqrt{3}} \sin \theta$
 $\frac{dx}{d\theta} = \frac{2}{\sqrt{3}} \cos \theta$
 $dx = \frac{2}{\sqrt{3}} \cos \theta d\theta$
 $x = 0 \Rightarrow \theta = 0$
 $x = 1 \Rightarrow \sin \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}$

12. $\int_1^{\sqrt{3}} \frac{x^2}{x^2+1} dx = \sqrt{3} - 1 - \frac{\pi}{12}$, use $x = \tan \theta$

Handwritten solution for problem 12:

$$\int_1^{\sqrt{3}} \frac{x^2}{x^2+1} dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan^2 \theta}{1+\tan^2 \theta} (\sec^2 \theta) d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan^2 \theta}{\sec^2 \theta} \sec^2 \theta d\theta = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \tan^2 \theta d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (\sec^2 \theta - 1) d\theta = \left[\tan \theta - \theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} = \left(\tan \frac{\pi}{3} - \frac{\pi}{3} \right) - \left(\tan \frac{\pi}{4} - \frac{\pi}{4} \right)$$

$$= \left(\sqrt{3} - \frac{\pi}{3} \right) - \left(1 - \frac{\pi}{4} \right) = \sqrt{3} - 1 - \frac{\pi}{12}$$

Substitution: $x = \tan \theta$
 $\frac{dx}{d\theta} = \sec^2 \theta$
 $dx = \sec^2 \theta d\theta$
 $x = 1 \Rightarrow \tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4}$
 $x = \sqrt{3} \Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}$

13. $\int_0^2 \sqrt{16-x^2} \, dx = \frac{1}{3}(4\pi + 6\sqrt{3})$, use $x = 4 \sin \theta$

Handwritten solution for problem 13:

$$\int_0^2 \sqrt{16-x^2} \, dx = \int_0^{\frac{\pi}{6}} \sqrt{16-(4\sin\theta)^2} \times 4\cos\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{6}} \sqrt{16-16\sin^2\theta} \times 4\cos\theta \, d\theta = \int_0^{\frac{\pi}{6}} 4\cos\theta \times 4\cos\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{6}} 16\cos^2\theta \, d\theta = \int_0^{\frac{\pi}{6}} 8(1+\cos 2\theta) \, d\theta = \int_0^{\frac{\pi}{6}} 8 \, d\theta + \int_0^{\frac{\pi}{6}} 8\cos 2\theta \, d\theta$$

$$= \left[8\theta + 4\sin 2\theta \right]_0^{\frac{\pi}{6}} = \left(\frac{4\pi}{3} + 2\sqrt{3} \right) - (0)$$

$$= \frac{1}{3}(4\pi + 6\sqrt{3})$$

Boxed notes:

- $x = 4\sin\theta$
- $\frac{dx}{d\theta} = 4\cos\theta$
- $x=0, \sin\theta=0 \Rightarrow \theta=0$
- $x=2, \sin\theta=\frac{1}{2} \Rightarrow \theta=\frac{\pi}{6}$

14. $\int_0^2 \frac{1}{(3x^2+4)^{\frac{3}{2}}} \, dx = \frac{1}{8}$, use $x = \frac{2}{\sqrt{3}} \tan \theta$

Handwritten solution for problem 14:

$$\int_0^2 \frac{1}{(3x^2+4)^{\frac{3}{2}}} \, dx = \dots$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{\left[3\left(\frac{2}{\sqrt{3}}\tan\theta\right)^2+4\right]^{\frac{3}{2}}} \times \frac{2}{\sqrt{3}} \sec^2\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{\left[4\tan^2\theta+4\right]^{\frac{3}{2}}} \times \frac{2}{\sqrt{3}} \sec^2\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{(4(\tan^2\theta+1))^{\frac{3}{2}}} \times \frac{2}{\sqrt{3}} \sec^2\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{(4\sec^2\theta)^{\frac{3}{2}}} \times \frac{2}{\sqrt{3}} \sec^2\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{8\sec^3\theta} \times \frac{2}{\sqrt{3}} \sec^2\theta \, d\theta = \int_0^{\frac{\pi}{6}} \frac{2\sec^2\theta}{8\sqrt{3}\sec^3\theta} \, d\theta = \int_0^{\frac{\pi}{6}} \frac{1}{4\sqrt{3}\sec\theta} \, d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{4\sqrt{3}} \cos\theta \, d\theta = \frac{1}{4\sqrt{3}} \left[\sin\theta \right]_0^{\frac{\pi}{6}} = \frac{1}{4\sqrt{3}} \left[\frac{\sqrt{3}}{2} - 0 \right] = \frac{1}{8}$$

Boxed notes:

- $x = \frac{2}{\sqrt{3}} \tan\theta$
- $\frac{dx}{d\theta} = \frac{2}{\sqrt{3}} \sec^2\theta$
- $x=0, \theta = \frac{2}{\sqrt{3}} \tan\theta \Rightarrow \tan\theta=0 \Rightarrow \theta=0$
- $x=2, 2 = \frac{2}{\sqrt{3}} \tan\theta \Rightarrow \tan\theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{6}$

15. $\int_0^2 \sqrt{16-3x^2} \, dx = \frac{8\pi\sqrt{3}}{9} + 2$, use $x = \frac{4}{\sqrt{3}} \sin \theta$

Handwritten solution for problem 15:

$$\int_0^2 \sqrt{16-3x^2} \, dx = \dots$$

$$= \int_0^{\frac{\pi}{6}} \sqrt{16-3\left(\frac{4}{\sqrt{3}}\sin\theta\right)^2} \times \frac{4}{\sqrt{3}} \cos\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{6}} \sqrt{16-16\sin^2\theta} \times \frac{4}{\sqrt{3}} \cos\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{6}} 4\cos\theta \times \frac{4}{\sqrt{3}} \cos\theta \, d\theta = \int_0^{\frac{\pi}{6}} \frac{16\cos^2\theta}{\sqrt{3}} \, d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{8}{\sqrt{3}} (1+\cos 2\theta) \, d\theta = \left[\frac{8}{\sqrt{3}}\theta + \frac{4}{\sqrt{3}}\sin 2\theta \right]_0^{\frac{\pi}{6}}$$

$$= \left(\frac{8}{\sqrt{3}} \times \frac{\pi}{6} + \frac{4}{\sqrt{3}} \times \frac{\sqrt{3}}{2} \right) - 0 = \frac{8\pi\sqrt{3}}{9} + 2$$

Boxed notes:

- $x = \frac{4}{\sqrt{3}} \sin\theta$
- $\frac{dx}{d\theta} = \frac{4}{\sqrt{3}} \cos\theta$
- $x=0 \rightarrow \theta=0$
- $x=2 \rightarrow \frac{4}{\sqrt{3}} = \frac{4}{\sqrt{3}} \sin\theta \Rightarrow \sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}$

16. $\int_0^3 \frac{27}{(9+x^2)^2} dx = \frac{\pi}{8} + \frac{1}{4}$, use $x = 3 \tan \theta$

Handwritten solution for the integral problem 16:

$$\int_0^3 \frac{27}{(9+x^2)^2} dx = \dots = \int_0^{\frac{\pi}{6}} \frac{27}{(9+9\tan^2\theta)^2} (3\sec^2\theta d\theta)$$

$$= \int_0^{\frac{\pi}{6}} \frac{27 \sec^2\theta}{9^2 (1+\tan^2\theta)^2} d\theta = \int_0^{\frac{\pi}{6}} \frac{27 \sec^2\theta}{81 \sec^4\theta} d\theta = \int_0^{\frac{\pi}{6}} \frac{1}{3 \sec^2\theta} d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{3} \cos^2\theta d\theta = \int_0^{\frac{\pi}{6}} \left(\frac{1}{6} + \frac{1}{6} \cos 2\theta \right) d\theta = \left(\frac{\theta}{6} + \frac{1}{12} \sin 2\theta \right) \Big|_0^{\frac{\pi}{6}} = \left(\frac{\pi}{36} + \frac{1}{12} \right) - (0+0)$$

$$= \frac{\pi}{36} + \frac{1}{12}$$

Boxed notes on the right:

$$\begin{cases} x = 3 \tan \theta \\ \frac{dx}{d\theta} = 3 \sec^2 \theta \\ dx = 3 \sec^2 \theta d\theta \\ x=0, \theta=0 \\ x=3, \theta=\frac{\pi}{6} \end{cases}$$